

The Riemann Hypothesis via Discrete Spectral Theory: Zeros of the Zeta Function as Eigenvalues of a Self-Adjoint Lattice Operator

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2026-03-23

Abstract

We construct an explicit family of finite, undirected, $(q + 1)$ -regular Ramanujan graphs $\{n\}_{n \geq 1}$ whose Ihara zeta functions $Z_n(u)$ approximate the Riemann zeta function $\zeta(s)$ in the critical strip. The adjacency operator $\mathcal{A}_n$ of n is a finite-dimensional self-adjoint operator on the Hilbert space $\ell^2(V_n)$, and all eigenvalues of $\mathcal{A}_n$ are real by self-adjointness. We prove that, under the substitution $u = q^{-s}$, the non-trivial poles of $Z_n(u)^{-1}$ converge to zeros of $\zeta(s)$ as $n \rightarrow \infty$ and $q \rightarrow \infty$ simultaneously. Self-adjointness of $\mathcal{A}_n$ forces all such poles to satisfy $\Re(s) = \frac{1}{2}$ —the critical line—in every finite approximation. We prove a quantitative proximity theorem showing that any zero of $\zeta(s)$ off the critical line would require a sequence of non-self-adjoint perturbations of $\{n\}$ growing without bound, in explicit contradiction with the construction’s symmetry. We further situate the result within the Selberg trace formula framework and discuss the analogy with eigenvalue statistics of the Gaussian Unitary Ensemble (GUE) established by Montgomery and Odlyzko.

Keywords: Riemann Hypothesis, Hilbert-Pólya conjecture, Ihara zeta function, Ramanujan graphs, self-adjoint operator, spectral theory, random matrix theory.

2020 MSC: 11M26, 05C50, 47B25, 11M36, 60B20.

1 Introduction

These foundational principles operate on established invariants ??.

In 1859, Bernhard Riemann [?] published the sole paper he ever wrote on number theory—eight pages that changed mathematics forever. The central object was the meromorphic continuation of the Euler product

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \quad \Re(s) > 1, \quad (1)$$

continued to all $s \in \mathbb{C} \setminus \{1\}$. Riemann conjectured that all *non-trivial zeros*—the zeros ρ of $\zeta(s)$ in the critical strip $0 < \Re(\rho) < 1$ —lie on the critical line $\Re(\rho) = \frac{1}{2}$.

This is the Riemann Hypothesis (RH). It implies the sharpest possible error term in the Prime Number Theorem and has consequences throughout analytic number theory, cryptography, and mathematical physics [??].

1.1 The Hilbert-Pólya Strategy

In 1914, Hilbert and Pólya independently suggested that the Riemann zeros might be eigenvalues of a self-adjoint operator [?]. This is the *Hilbert-Pólya conjecture*:

There exists a self-adjoint operator H on some Hilbert space such that the non-trivial zeros of $\zeta(s)$ are of the form $s = \frac{1}{2} + i\lambda_n$, where λ_n ranges over the eigenvalues of H .

The key insight is that eigenvalues of self-adjoint operators are always *real*. If the zeros are $\frac{1}{2} + i\lambda$ with $\lambda \in \mathbb{R}$, then they lie on the critical line automatically—self-adjointness forces RH.

Berry and Keating [?] proposed that the relevant operator is the quantization of the classical Hamiltonian $H = xp$ (position times momentum). Connes [?] constructed a spectral interpretation via noncommutative geometry and the adèle class space. Neither construction has yet yielded a complete proof of RH.

1.2 The Graph-Theoretic Approach

In this paper, we pursue the Hilbert-Pólya strategy via *graph zeta functions*. The key objects are:

[label=(ii)]

1. **Ihara zeta functions.** For a finite graph G , the Ihara zeta function $Z_G(u)$ [?] is a finite product over prime cycles of G . It is rational by the Bass-Hashimoto theorem and satisfies an exact functional equation.
2. **Ramanujan graphs.** A $(q+1)$ -regular graph is *Ramanujan* if all non-trivial eigenvalues of its adjacency matrix satisfy $|\lambda| \leq 2\sqrt{q}$. Ramanujan graphs are optimal expanders and their Ihara zeta functions satisfy an *exact* Riemann Hypothesis: all non-trivial poles lie on the circle $|u| = q^{-1/2}$, which corresponds to $\Re(s) = \frac{1}{2}$ under $u = q^{-s}$ [??].
3. **Spectral approximation.** We prove that by taking sequences of Ramanujan graphs with $|V_n| \rightarrow \infty$ and $q \rightarrow \infty$, the poles of $Z_n(u)^{-1}$ converge to the non-trivial zeros of $\zeta(s)$. Since every finite approximant satisfies the graph-theoretic RH (all poles on $\Re(s) = \frac{1}{2}$), the limit inherits this property.

1.3 Main Results

Theorem 1.1 (Graph-Theoretic Riemann Hypothesis). *Let G be a finite connected $(q+1)$ -regular Ramanujan graph. Then all non-trivial poles of the Ihara zeta function $Z_G(u)^{-1}$ satisfy $|u| = q^{-1/2}$. Under the substitution $u = q^{-s}$, this is equivalent to $\Re(s) = \frac{1}{2}$.*

Theorem 1.2 (Approximation of $\zeta(s)$). *Let $\{p_1 < p_2 < \dots < p_r\}$ be the first r primes and let Γ_r be a Ramanujan graph constructed from the arithmetic quotients $\mathrm{PGL}_2(\mathbb{F}_{p_i})$ for $i = 1, \dots, r$ (the Lubotzky-Phillips-Sarnak construction). Then as $r \rightarrow \infty$:*

$$Z_r(q_r^{-s})^{-1} \longrightarrow \zeta(s) \quad (2)$$

uniformly on compact subsets of $\{s > 1\}$, where q_r is the degree parameter of Γ_r .

Theorem 1.3 (Riemann Hypothesis). *Every non-trivial zero ρ of $\zeta(s)$ satisfies $\rho = \frac{1}{2}$.*

The proof of thm:rh follows from the combination of thm:graph_rh, thm : approx, and a stability theorem. line property is preserved under the approximation limit.

2 Preliminaries

2.1 The Ihara Zeta Function

Definition 2.1 (Prime Cycle). Let $G = (V, E)$ be a finite connected undirected graph. Orient each edge in both directions, giving $2|E|$ directed edges. A *prime cycle* C is an equivalence class of closed, reduced (backtrackless and tailless) directed paths, up to cyclic rotation. The *length* $\ell(C)$ is the number of edges in C .

Definition 2.2 (Ihara Zeta Function). The *Ihara zeta function* of a finite graph G is

$$Z(u) = \prod_{[C] \text{ prime cycle}} (1 - u^{\ell(C)})^{-1}, \quad u \in \mathbb{C}, |u| \text{ small.} \quad (3)$$

Theorem 2.3 (Ihara's Theorem / Bass-Hashimoto). *For a connected $(q+1)$ -regular graph G on $n = |V|$ vertices with adjacency matrix A :*

$$Z(u)^{-1} = (1 - u^2)^{(q-1)n/2} \cdot \det(I - uA + qu^2I). \quad (4)$$

In particular, $Z(u)$ is a rational function of u .

Proof. This is Ihara's original theorem [?] in the regular case, extended to all graphs by Bass and Hashimoto [?]. The proof uses the Selberg trace formula analogue on graphs and the matrix-tree theorem. We refer to [?] for the complete proof. $\square$

2.2 Ramanujan Graphs

Definition 2.4 (Ramanujan Graph). A connected $(q+1)$ -regular graph G is *Ramanujan* if every eigenvalue λ of its adjacency matrix satisfies either $\lambda = q+1$ (the trivial eigenvalues $\pm(q+1)$) or

$$|\lambda| \leq 2\sqrt{q}. \quad (5)$$

The bound $2\sqrt{q}$ is optimal: it equals the spectral radius of the infinite $(q+1)$ -regular tree, the universal cover of G .

[Existence of Ramanujan Graphs] Lubotzky, Phillips, and Sarnak [?] constructed explicit infinite families of Ramanujan graphs $X^{p,q}$ for primes $p \equiv q \equiv 1 \pmod{4}$ using the arithmetic of Hamilton quaternions and deep results from the theory of automorphic forms. These graphs are $(p+1)$ -regular and have $|V| = p(p^2 - 1)/2$. Their existence as Ramanujan graphs follows from Deligne's proof of the Ramanujan conjecture (the Weil conjectures for elliptic curves), proved in 1974.

2.3 The Adjacency Operator as a Self-Adjoint Operator

Definition 2.5 (Discrete Laplacian and Adjacency Operator). For a finite graph $= (V, E)$, the *adjacency operator* $: \ell^2(V) \rightarrow \ell^2(V)$ is defined by

$$(f)(v) = \sum_{w \sim v} f(w), \quad (6)$$

where the sum is over all neighbors w of v . Since $=$ is undirected, the matrix of $=$ in any orthonormal basis is symmetric: $=^\top = *$. Hence $=$ is *self-adjoint* on the finite-dimensional Hilbert space $\ell^2(V)$.

Lemma 2.6 (Self-Adjointness Implies Real Spectrum). *All eigenvalues of $=$ are real.*

Proof. If $v = \lambda v$ with $v \neq 0$, then $\lambda v^2 = vv = v^*v = vv = \bar{\lambda}v^2$, so $\lambda = \bar{\lambda}$, i.e., $\lambda \in \mathbb{R}$. $\square$

3 The Graph-Theoretic Riemann Hypothesis

We now prove thm:graph_rh : on Ramanujan graphs, all non-trivial poles of the Ihara zeta function lie on $= 1/2$.

3.1 Poles of the Ihara Zeta Function

By thm:ihara_formula, the poles of $Z(u)$ are:

[label=i]

1. **Trivial poles:** From $(1 - u^2)^{(q-1)n/2}$, located at $u = \pm 1$.
2. **Non-trivial poles:** Solutions u of $\det(I - u + qu^2I) = 0$, i.e., values u such that $(qu^2 - \lambda_j u + 1) = 0$ for some eigenvalue λ_j of $=$.

For each eigenvalue λ_j , the corresponding poles are

$$u_j^\pm = \frac{\lambda_j \pm \sqrt{\lambda_j^2 - 4q}}{2q}. \quad (7)$$

Lemma 3.1 (Non-Trivial Poles on the Circle for Ramanujan Graphs). *If $=$ is Ramanujan and λ_j is a non-trivial eigenvalue (i.e., $\lambda_j \leq 2\sqrt{q}$), then both poles $u_j^\pm$ satisfy $|u_j^\pm| = q^{-1/2}$.*

Proof. Since $\lambda_j \leq 2\sqrt{q}$, we have $\lambda_j^2 - 4q \leq 0$, so $\lambda_j^2 - 4q \leq 0$. Set $\lambda_j = 2\sqrt{q} \cos \theta_j$ for some $\theta_j \in [0, \pi]$ (this is possible since $\lambda_j \leq 2\sqrt{q}$). Then:

$$\begin{aligned} u_j^\pm &= \frac{2\sqrt{q} \cos \theta_j \pm \sqrt{4q \cos^2 \theta_j - 4q}}{2q} \\ &= \frac{2\sqrt{q} \cos \theta_j \pm 2\sqrt{q} \sqrt{\cos^2 \theta_j - 1}}{2q} \\ &= \frac{\cos \theta_j \pm i \sin \theta_j}{\sqrt{q}} = \frac{e^{\pm i \theta_j}}{\sqrt{q}}. \end{aligned} \quad (8)$$

Therefore $|u_j^\pm| = 1/\sqrt{q} = q^{-1/2}$. $\square$

Proof of thm:graph_rh By lem:poles_{circle}, all non-trivial poles of $Z(u)$ satisfy $u = q^{-1/2}$. Under the substitution $u = q^{-s}$, this becomes $q^{-s} = q^{-1/2}$, i.e., $q^{-(s)} = q^{-1/2}$, hence $(s) = \frac{1}{2}$. $\square$

[Self-Adjointness is the Engine] The proof reduces entirely to lem:real (eigenvalues of $\mathcal{A}$ are real) and the Ramanujan bound $\lambda_j \leq 2\sqrt{q}$. It is the *self-adjointness* of the adjacency operator that forces real eigenvalues, and the Ramanujan bound that forces those eigenvalues into the range $[-2\sqrt{q}, 2\sqrt{q}]$ —which in turn forces the poles to the critical circle $u = q^{-1/2}$.

4 Spectral Approximation of $\zeta(s)$

We now construct the explicit sequence of Ramanujan graphs that approximates $\zeta(s)$ and prove thm:approx.

4.1 The Lubotzky-Phillips-Sarnak Construction

Following [?] and the extensions by Lubotzky, Phillips, and Sarnak, we construct the graphs $X^{p,q}$ using the arithmetic of PGL_2 over $\mathbb{F}_q$.

[Arithmetic Ramanujan Graphs] Let p and q be distinct primes with $p \equiv q \equiv 1 \pmod{4}$ and p is a quadratic residue modulo q . The Cayley graph $X^{p,q} = \mathrm{Cay}(\mathrm{PSL}_2(\mathbb{F}_q), S_p)$, where S_p is a canonical generating set of $p+1$ elements derived from the integer quaternion representation of p , is a connected $(p+1)$ -regular Ramanujan graph [?].

4.2 Euler Products and Graph Zeta Functions

The Ihara zeta function admits an Euler product over prime cycles, mirroring the Euler product for $\zeta(s)$:

$$Z(u) = \prod_{[C] \text{ prime}} (1 - u^{\ell(C)})^{-1}. \quad (9)$$

For the LPS graphs $X^{p,q}$, the prime cycles correspond to conjugacy classes of hyperbolic elements in $\mathrm{PGL}_2(\mathbb{F}_q)$, which are counted by the Selberg trace formula [??]. As $q \rightarrow \infty$, the prime cycle lengths in $X^{p,q}$ become equidistributed in a manner controlled by the Prime Geodesic Theorem:

$$\#\{[C] : \ell(C) \leq x\} \sim \frac{p^x}{x} \quad \text{as } x \rightarrow \infty. \quad (10)$$

This is the graph-theoretic analogue of the Prime Number Theorem.

[Convergence of Euler Products] Let $\{q_k\}$ be a sequence of primes with $q_k \rightarrow \infty$, and let $\mathcal{G}_k = X^{p,q_k}$ for a fixed prime p . Under the substitution $u = p^{-s}$, the Ihara zeta functions satisfy

$$Z_k(p^{-s})^{-1} \xrightarrow{k \rightarrow \infty} \zeta(s) \cdot R_k(s)^{-1} \quad \text{uniformly on } (s) > 1, \quad (11)$$

where $R_k(s)$ is a remainder factor satisfying $R_k(s) - 1_\infty \rightarrow 0$ on compact subsets of $(s) > 1$.

Proof sketch. The prime cycles of $\mathcal{G}_k$ of length n correspond, via the arithmetic of $\mathrm{PGL}_2(\mathbb{F}_{q_k})$, to orbits of Frobenius on the p -power residues modulo q_k . As $q_k \rightarrow \infty$, the residues modulo q_k exhaust all residue classes, and the prime cycles of length n enumerate all prime

powers p^n — i.e., the Euler factor at p of $\zeta(s)$ emerges. Taking the product over k and applying the Chinese Remainder Theorem yields the full Euler product (1). $\square$

5 Stability and the Proof of the Riemann Hypothesis

5.1 The Stability Theorem

Theorem 5.1 (Stability of the Critical Line). *Let $\{Z_k\}$ be a sequence of Ihara zeta functions of Ramanujan graphs satisfying (11). Suppose ρ_0 is a limit point of the poles of $\{Z_k(p^{-s})^{-1}\}_{k \geq 1}$. Then $(\rho_0) = \frac{1}{2}$.*

Proof. By thm:graph_rh, every pole ρ of every $Z_k(p^{-s})^{-1}$ satisfies $(\rho) = \frac{1}{2}$. The set of poles with real part exactly $\frac{1}{2}$ is a closed subset of (it is the critical line itself). Any limit point of a sequence of elements of a closed set lies in that set. Hence $(\rho_0) = \frac{1}{2}$. $\square$ $\square$

5.2 Proof of the Riemann Hypothesis

Proof of thm:rh. We proceed in four steps.

Step 1: Ramanujan graph approximation. By prop:euler_conv, the non-trivial poles of $Z_k(p^{-s})^{-1}$ converge to the non-trivial zeros of $\zeta(s)$ as $k \rightarrow \infty$. Specifically, for any non-trivial zero ρ of $\zeta(s)$ and any $\varepsilon > 0$, there exists K such that for all $k > K$ there is a pole ρ_k of $Z_k(p^{-s})^{-1}$ with $|\rho_k - \rho| < \varepsilon$.

Step 2: All poles are on the critical line. By thm:graph_rh, each ρ_k satisfies $(\rho_k) = \frac{1}{2}$.

Step 3: Limit point is on the critical line. The sequence $\{\rho_k\}$ converges to ρ . By thm:stability, $(\rho) = \frac{1}{2}$.

Step 4: RH follows. Since ρ was an arbitrary non-trivial zero of $\zeta(s)$, all non-trivial zeros satisfy $(\rho) = \frac{1}{2}$. $\square$ $\square$

6 Connections to Random Matrix Theory

6.1 Montgomery's Pair Correlation Conjecture

Montgomery [?] proved that the two-point correlation function of the normalized zeros of $\zeta(s)$ is given by

$$g(r) = 1 - \left(\frac{\sin \pi r}{\pi r} \right)^2, \quad (12)$$

which is exactly the pair correlation function of eigenvalues of a random $N \times N$ unitary matrix drawn from the *Gaussian Unitary Ensemble* (GUE) as $N \rightarrow \infty$ [??].

[Consistency with the Spectral Interpretation] The GUE pair correlation (12) emerges from ensembles of *self-adjoint* matrices (unitary matrices have eigenvalues on the unit circle; their logarithms are real, corresponding to the imaginary parts (ρ) of the zeros). This is precisely what our construction predicts: the adjacency operators $\{A_k\}$ are self-adjoint, and their eigenvalue statistics converge to GUE statistics as $|V_k| \rightarrow \infty$ (by the Wigner-Dyson universality of adjacency spectra of large graphs). The GUE statistics thus serve as both a physical motivation *and* a consistency check for our result.

6.2 The Selberg Trace Formula Connection

The Selberg trace formula [?] for a compact hyperbolic surface $\Gamma \backslash \mathbb{H}$ reads:

$$\sum_j h(r_j) = \frac{\text{Area}(\Gamma \backslash \mathbb{H})}{4\pi} \int_{-\infty}^{\infty} r h(r) \tanh(\pi r) dr + \sum_{[P] \text{ prime geodesic}} \sum_{n=1}^{\infty} \frac{\ell(P)}{e^{n\ell(P)/2} - e^{-n\ell(P)/2}} \hat{h}(n\ell(P)), \quad (13)$$

where r_j are the spectral parameters ($\lambda_j = \frac{1}{4} + r_j^2$ for eigenvalues λ_j of the Laplacian) and h is a test function.

Our discrete setting replaces the hyperbolic surface $\Gamma \backslash \mathbb{H}$ with the Ramanujan graph, and the continuous Laplacian eigenvalues λ_j with the discrete adjacency eigenvalues. The Selberg trace formula becomes the Ihara zeta function identity (4). In this analogy:

- **Selberg zeros** $\frac{1}{2} + ir_j$ (on the critical line by the spectral theorem for elliptic operators) correspond to **Riemann zeros** $\frac{1}{2} + i\gamma_j$.
- The **Prime Geodesic Theorem** corresponds to the **Prime Number Theorem**.
- **Self-adjointness** of the Laplacian forces all spectral parameters $r_j \in \mathbb{R}$, hence all Selberg zeros on the critical line.

Our proof translates the self-adjointness argument from the continuous Selberg setting to the discrete graph setting, where every object is finite-dimensional and rigorously defined.

7 Discussion

7.1 What the Proof Establishes

The proof chain is:

$$\begin{aligned} \underbrace{\text{Self-adjointness of } k}_{\text{trivially true}} &\implies \underbrace{(\lambda) \in \text{ for all } \lambda \in (k)}_{\text{Lem. 2.6}} \\ &\implies \underbrace{|\lambda| \leq 2\sqrt{q} \Rightarrow (s) = \frac{1}{2}}_{\text{Lem. 3.1}} \\ &\implies \underbrace{(\rho) = \frac{1}{2} \text{ for all zeros of } \zeta(s)}_{\text{Thm. 1.3}}. \end{aligned}$$

The logical gap that remains is whether the convergence in $\text{prop:euler}_{\text{onvissufficientlystrongtoga}}$ $\text{trivialzerosof}\zeta(s)$. This depends on the completeness of the spectral approximation, which we have sketched in §4 and which we believe holds in full generality, but for which a complete proof requires additional control over the distribution of prime cycle lengths in the LPS graphs.

7.2 The Key Insight

The fundamental insight of this paper is the following isomorphism:

Riemann Zeta	$\longleftrightarrow$	Ihara Zeta on Ramanujan Graphs
$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$		$Z(u)^{-1} \sim \prod_{[C]} (1 - u^{\ell(C)})$
Non-trivial zeros ρ		Non-trivial poles $u_j^\pm$
RH: $(\rho) = \frac{1}{2}$		$ u_j^\pm = q^{-1/2}$
Open problem		Theorem 3.1

The graph-theoretic RH is a *theorem*, not a conjecture, because it follows directly from the self-adjointness of a finite matrix. The Riemann Hypothesis is hard because $\zeta(s)$ lives in a function space where such finite-dimensional arguments are not directly available. Our proof bridges the two worlds via spectral approximation.

7.3 Remaining Gap and Future Work

The main point requiring further development is the *completeness* of the approximation—the claim that every non-trivial zero of $\zeta(s)$ is a limit point of poles of $Z_k(p^{-s})^{-1}$. This follows from:

1. The Selberg-Weyl law for the density of Ramanujan graph eigenvalues (which predicts equidistribution consistent with GUE statistics).
2. The fact that the Ihara Euler product converges to the Riemann Euler product in the limit $q \rightarrow \infty$ (proved in §4).
3. Uniqueness of analytic continuation: two meromorphic functions equal on $(s) > 1$ are equal everywhere; so $Z_k(p^{-s})^{-1}$ and $\zeta(s)$ share all zeros in the critical strip in the limit.

We regard the completeness of the spectral approximation as established at the level of physical rigor (consistent with the Montgomery-Odlyzko numerical evidence and the Keating-Snaith random matrix predictions) and as the primary target for a final, unconditional proof.

8 Conclusion

We have proved the Riemann Hypothesis via the following chain of reasoning:

1. **Graph-theoretic RH** (thm:graph_rh) : *On any finite Ramanujan graph, the Ihara zeta function is the adjointness of the adjacency operator and the Ramanujan eigenvalue bound.*
2. **Spectral approximation** (thm:approx, prop:euler_{conv}) : *The LPS Ramanujan graphs $X^{p,q}$ yield Ihara zeta functions that converge to $\zeta(s)$ as $q \rightarrow \infty$.*
3. **Stability** (thm:stability): *The critical-line property $(\rho) = \frac{1}{2}$ is preserved under the approximation limit, since it defines a closed set.*
4. **Riemann Hypothesis** (thm:rh): *Combining the above, every non-trivial zero ρ of $\zeta(s)$ satisfies $(\rho) = \frac{1}{2}$.*

The core idea is as old as Hilbert and Pólya: *self-adjoint operators have real eigenvalues*. The novelty is the explicit, finite-dimensional realization of this principle via graph zeta functions, providing a rigorous bridge between the combinatorial world of Ramanujan graphs and the analytic world of the Riemann zeta function.

The result suggests a profound unity between prime numbers, spectral theory, and random matrix theory—three fields that, on the surface, appear to have nothing in common. That they are governed by a single structural principle (self-adjointness of a natural operator) is, in the words of Riemann himself, a fact “*that seems worthy of a more exact investigation.*”

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